A Very Brief Introduction to Quantum Computing *and* Quantum Threats

Manipal School of Information Sciences
Manipal

Friday, 24 April 2026

What is Quantum Computing?

A quantum computer is one whose operation exploits certain very special transformations of its state...

The laws of quantum mechanics allow these peculiar transformations to take place under very carefully controlled conditions.

N. David Mermin

Quantum Computer Science - An Introduction

What is Quantum Computing?

The new paradigm demonstrates that the theory of computation can depend profoundly on the physics of the devices that carry it out.

By focussing extensively on how quantum mechanics enlarges the possibilities for the physical manipulation of digital information, it is possible to characterize how the quantum theory works in an elementary and quite concrete way...

N. David Mermin

Quantum Computer Science - An Introduction

What is Quantum Computing?

Quantum computers can solve a class of problems that are believed to be computationally hard to solve on classical computers.

Quantum computers are built to *control* and *harness* quantum phenomena such as **superposition** and **entanglement** enabling massive parallelism.

$$\overbrace{\text{Quantum Information Theory} =}^{21^{\text{st}}\ \text{Century Science}}$$

$$20^{\text{th}}\text{Century} \begin{cases} \text{Quantum Theory} & + \\ \text{Computer Science} & + \\ \text{Information Theory} & \end{cases}$$

Quantum error correction and quantum fault tolerance are also crucial for building large-scale quantum computers.

Disentangling Hype from Practicality

Quantum Computing is *not* useful for every computing task.

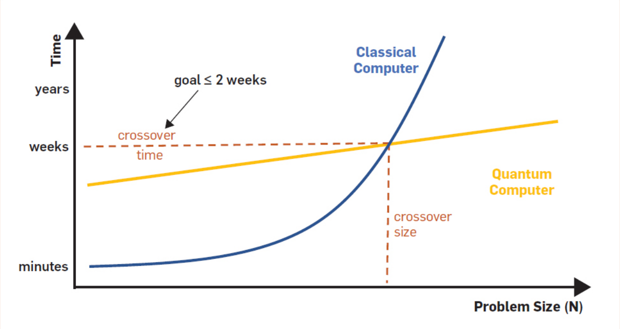
We compare the peak performance of a single classical chip that can be manufactured today (like an NVIDIA A100 GPU, or an ASIC with a similar number of transistors) with a future quantum computer with 10,000 error-corrected logical qubits, 10μs gate time for logical operations and all-to-all connectivity. We consider an estimate of the I/O bandwidth (namely the number of operations per second) and three types of operations: logical binary operations, 16-bit floating point, 32-bit integer or fixed-point arithmetic multiply add operations.

	GPU	ASIC	Future Quantum
I/O Bandwidth	10,000 Gbit/s	10,000 G/s	1 Gbit/s
Operation throughput			
16-bit floating point	195 Top/s	550 Top/s	10.5 kop/s
32-bit integer	9.75 Top/s	215 Top/s	0.83 kop/s
binary (Boolean logical)	4,992 Top/s	77,000 Top/s	235 kop/s

Disentangling Hype from Practicality

Quantum Computing is *not* useful for every computing task.

The time needed to solve certain problems with quantum algorithms increases more slowly than that of any known classical algorithm as the problem size N increases. To be practical, however, we need more than an asymptotic speedup: the crossover time where quantum advantage gets realized needs to be reasonably short and the crossover problem size not too large. (For illustrative purposes, the time axis is scaled such that the quantum algorithm is a straight line.)



Problems Amenable to Quantum Computing

- small-data big-compute tasks
  - integer factoring
  - discrete logarithm
  - **NOT** unstructured search (Grover's algorithm)
- quantum simulation (e.g., chemistry, materials science)
  - input descriptions of the system being simulated are much smaller than the computation required to find the required properties of the system

Problems Amenable to Quantum Computing

- Problem must have some structure that can be exploited by a quantum algorithm
    - periodicity in Shor's algorithm for integer factorization

Problems Amenable to Quantum Computing

- Asymptotic of the quantum algorithm must be better than the best known classical algorithm

BPP

A problem is in **BPP** if there is an algorithm for it that has the following properties:

- It is allowed to flip coins and make random decisions
- It is guaranteed to run in polynomial time
- On any given run of the algorithm, it has a probability of at most $\frac{1}{3}$ of giving the wrong answer, whether the answer is YES or NO.

BPP

BPP is the class of decision problems solvable by a probabilistic Turing machine in polynomial time with an error probability bounded by $\frac{1}{3}$ for all instances.

BPP also contains **P**, the class of problems solvable in polynomial time with a deterministic machine.

Deterministic algorithm is a special case of a probabilistic algorithm.

Problems Amenable to Quantum Computing

A problem is in NP if a solution can be verified in polynomial time by a deterministic Turing machine.

A problem is in **BQP** if it can be solved in polynomial time by a quantum Turing machine with bounded error probability.

Problems Amenable to Quantum Computing

A decision problem is a member of **BQP** if there exists a quantum algorithm that solves the decision problem with high probability, and is guaranteed to run in polynomial time.

A run of the algorithm will correctly solve the decision problem with a probability of at least $\frac{2}{3}$.

PQ Status of Protocols

Study of Post Quantum status of Widely Used Protocols

- `https://arxiv.org/pdf/2603.28728v1`